BME lecture week:2

Textbooks:

[1] R.S. Khurmi & J.K.Gupta, “A Text Book of Thermal Engineering”, two colour edition 1978(Reprint 2003), S.Chand & Company Ltd.

[2] R.S. Khurmi, “A Text Book of Hydraulics, Fluid Mechanics and Hydraulic Machines”, S.Chand & Company Ltd, Reprint 2007

References:

[1] V.P.Vasandani & D.S. Kumar, “Heat Engineering”, Metropolitan Book Co.(Pvt.) Ltd. Reprint 2001

[2] M.J. Moran & H.N. Shapiro, “Engineering Thermodynamics”, John Wiley and Sons.2000

Course contents of first 2 weeks

Week1: *Properties of steam – internal energy, enthalpy, entropy and quality of steam, saturated and superheated steam, uses of steam table and Mollier diagram*

Week2: *Steam generating unit, boiler mountings &accessories*.

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This lecture will cover the theory of steam boilers; boiler mounting and accessories; and performance of steam boilers

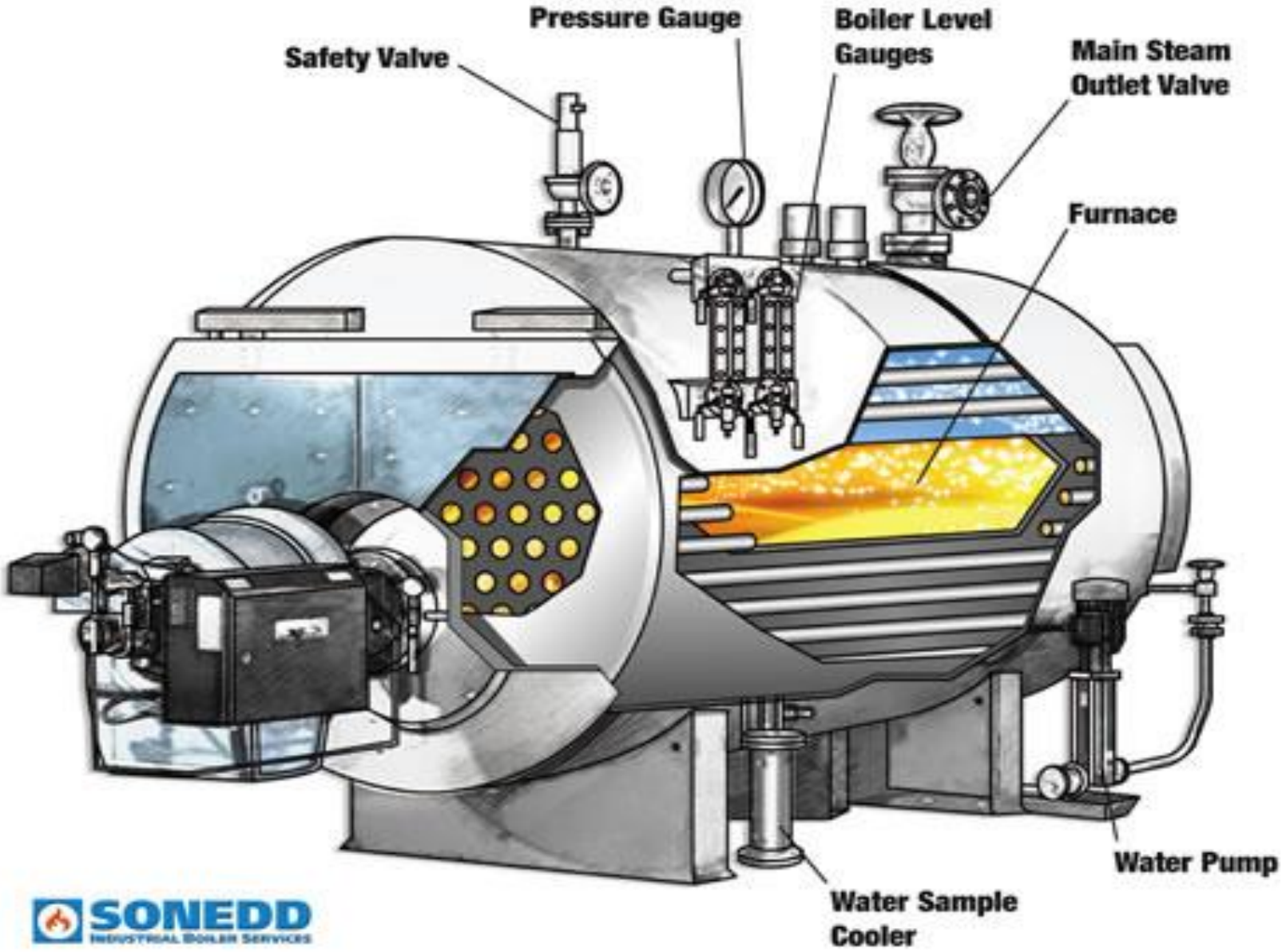
Steam boilers

13.1. Introduction

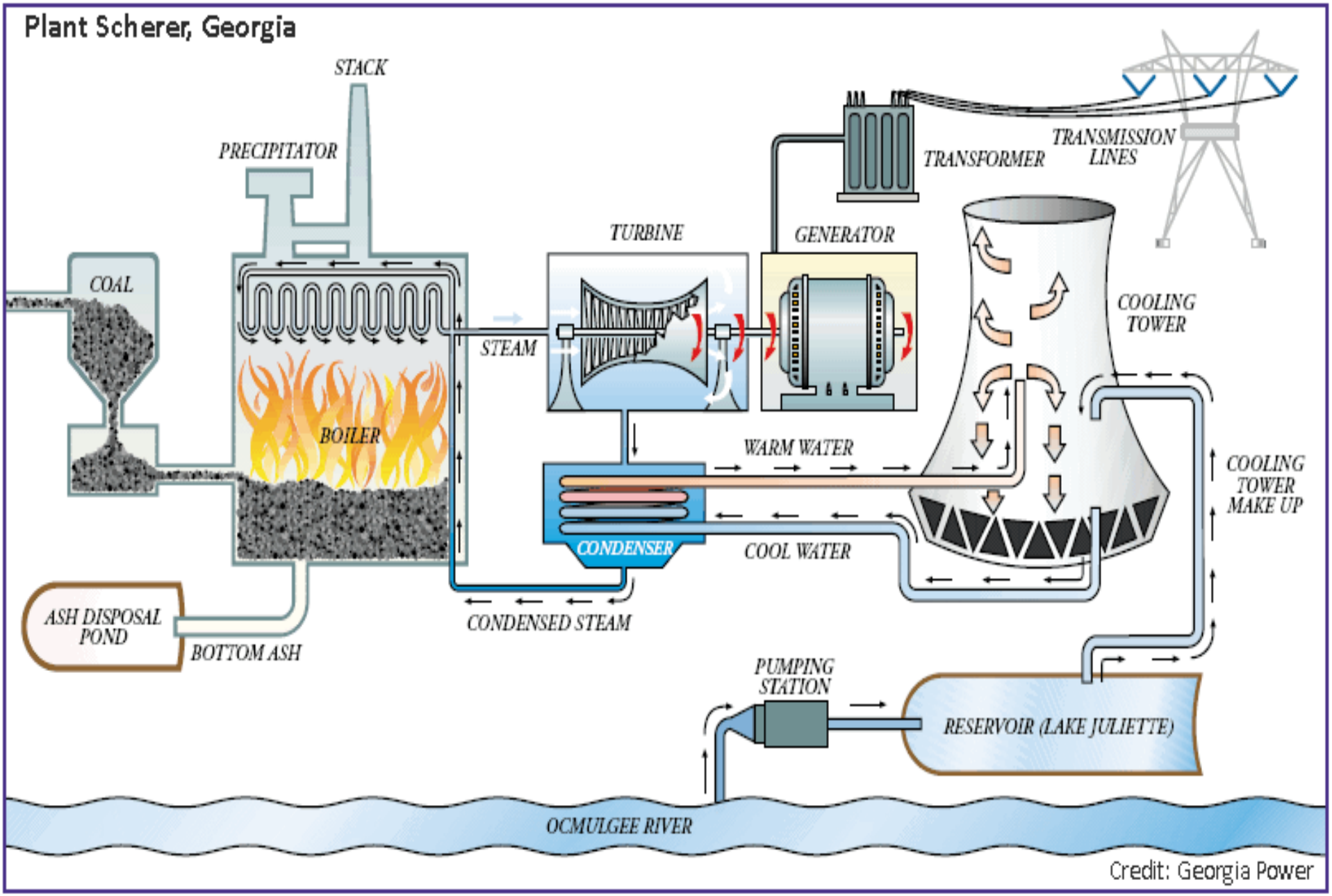
A steam generator or boiler is, usually, a closed vessel made of steel. Its function is to transfer the heat produced by the combustion of fuel (solid, liquid or gaseous) to water, and ultimately to generate steam. The steam produced may be supplied

1. to an external combustion engine, *i.e.* steam engines and turbines,
2. at low pressures for industrial process work in cotton mills, sugar factories, breweries, etc., and
3. for producing hot water, which can be used for heating installations at much lower pressures.

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13.2. Important Terms for Steam Boilers

Though there are many terms used in steam boilers, yet the following are important from the subject point of view

1. *Boiler shell.* It is made up of steel plates bent into cylindrical form and riveted or welded together. The ends of the shell are closed by means of end plates. A boiler shell should have sufficient capacity to contain water and steam.

2. *Combustion chamber.* It is the space, generally below the boiler shell, meant for burning fuel in order to produce steam from the water contained in the shell.

3. *Grate.* It is a platform, in the combustion chamber, upon which fuel (coal or wood) is burnt. The grate, generally, consists of cast iron bars which are spaced apart so that air (required for combustion) can pass through them. The surface area of the grate, over which the fire takes place, is called *grate surface*.

4. *Furnace.* It is the space, above the grate and below the boiler shell, in which the fuel is actually burnt. The furnace is also called *fire box*.

5. *Heating surface.* It is that part of boiler surface, which is exposed to the fire (or hot gases from the fire).

6. *Mountings.* These are the fittings which are mounted on the boiler for its proper functioning. They include water level indicator, pressure gauge, safety valve etc. It may be noted that a boiler cannot function safely without the mountings.

7. *Accessories.* These are the devices, which form an integral part of a boiler, but are not mounted on it. They include superheater, economiser, feed pump etc. It may be noted that the accessories help in controlling and running the boiler efficiently.

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13.5. Classifications of Steam Boilers

Though there are many classifications of steam boilers, yet the following are important from the subject point of view :

1. *According to the contents in the tube.* The steam boilers, according to the contents in the tube may be classified as :

(*a*) Fire tube or smoke tube boiler, and (*b*) Water tube boiler.

2. *According to the position of the furnace.* The steam boilers, according to the position of the furnace are classified as :

(*a*) Internally fired boilers, and (*b*) Externally fired boilers.

3. *According to the axis of the shell.* The steam boilers, according to the axis of the shell, may be classified as :

(*a*) Vertical boilers, and (*b*) Horizontal boilers.

4. *According to the number of tubes.* The steam boilers, according to the number of tubes, may be classified as :

(*a*) Single tube boilers, and (*b*) Multitubular boilers.

5. *According to the method of circulation of water and steam.* The steam boilers, according to the method of circulation of water and steam, may be classified as :

(*a*) Natural circulation boilers, and (*b*) Forced circulation boilers.

6. *According to the use.* The steam boilers, according to their use, may be classified as

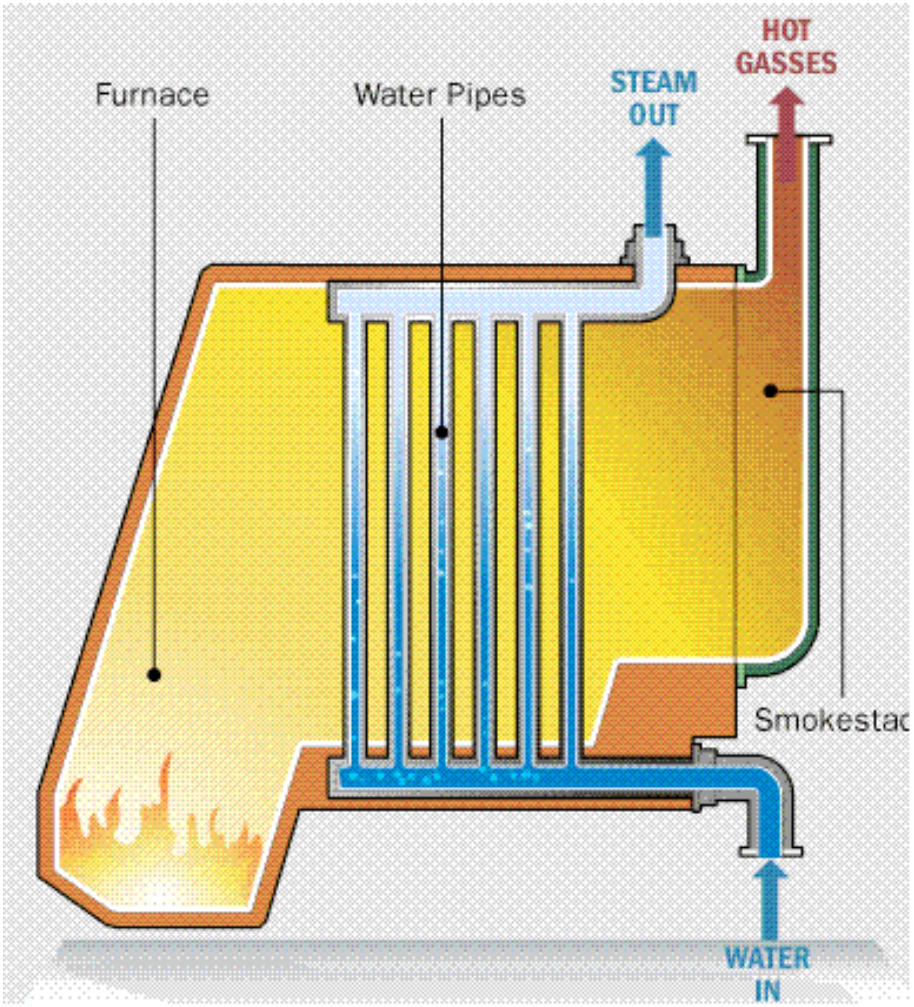
(*a*) Stationary boilers, and (*b*) Mobile boilers

13.16. Comparison Between Water Tube and Fire Tube Boilers

Following are the few points of comparison between a water tube and a fire tube boiler.

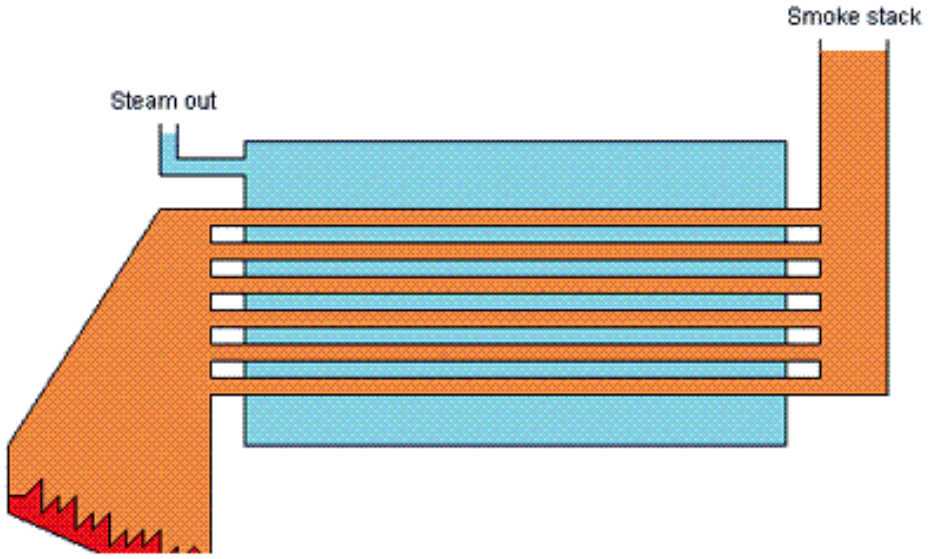
S.No.	*Water tube boiler*	*Fire tube boiler*
1.	The water circulates inside the tubes which are surrounded by hot gases from the furnace	The hot gases from the furnace pass through the tubes which are surrounded by water
2.	It generates steam at a higher pressure upto 165 bar.	It can generate steam only upto 24.5 bar
3.	The rate of generation of steam is high, *i.e.* upto 450 tonnes per hour.	The rate of generation of steam is low, *i.e.* upto 9 tonnes per hour
4.	For a given power, the floor area required for the generation of steam is less, *i.e.* about 5 m^2 per tonne per hour of steam generation.	The floor area required is more, *i.e.* about 8 m^2 per tonne per hour of steam generation
5.	Overall efficiency with economiser is upto 90%	Its overall efficiency is only 75%
6.	It can be transported and erected easily as its various parts can be separated.	The transportation and erection is difficult.
7.	It is preferred for widely fluctuating loads.	It can also cope reasonably with sudden increase in load but for a shorter period
8.	The direction of water circulation is well defined	The water does not circulate in a definite direction
9.	The operating cost is high	The operating cost is less
10.	The bursting chances are more.	The bursting chances are less
11.	The bursting does not produce any destruction to the whole boiler.	The bursting produces greater risk to the damage of the property
12.	It is used for large power plants.	It is not suitable for large plants

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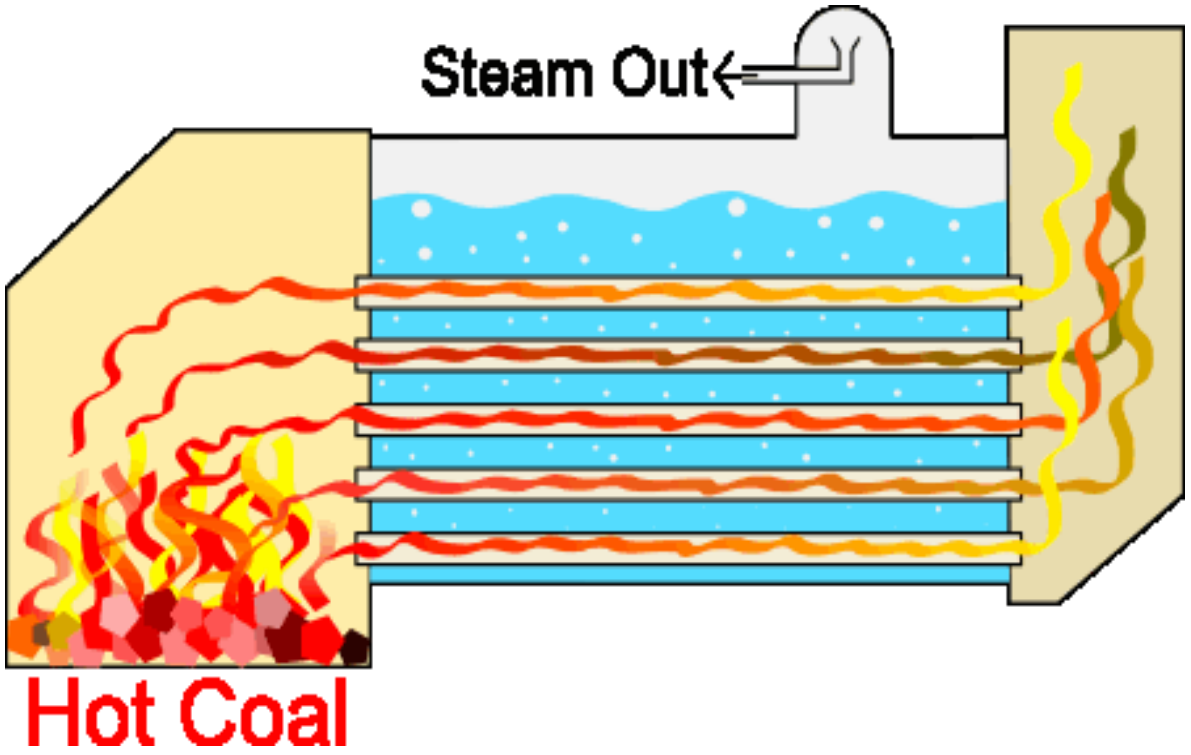


water tube boiler

Difference between fire tube and water tube boiler



fire tube boiler


Steam Out
Hot Coal

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Boiler mountings and Accessories

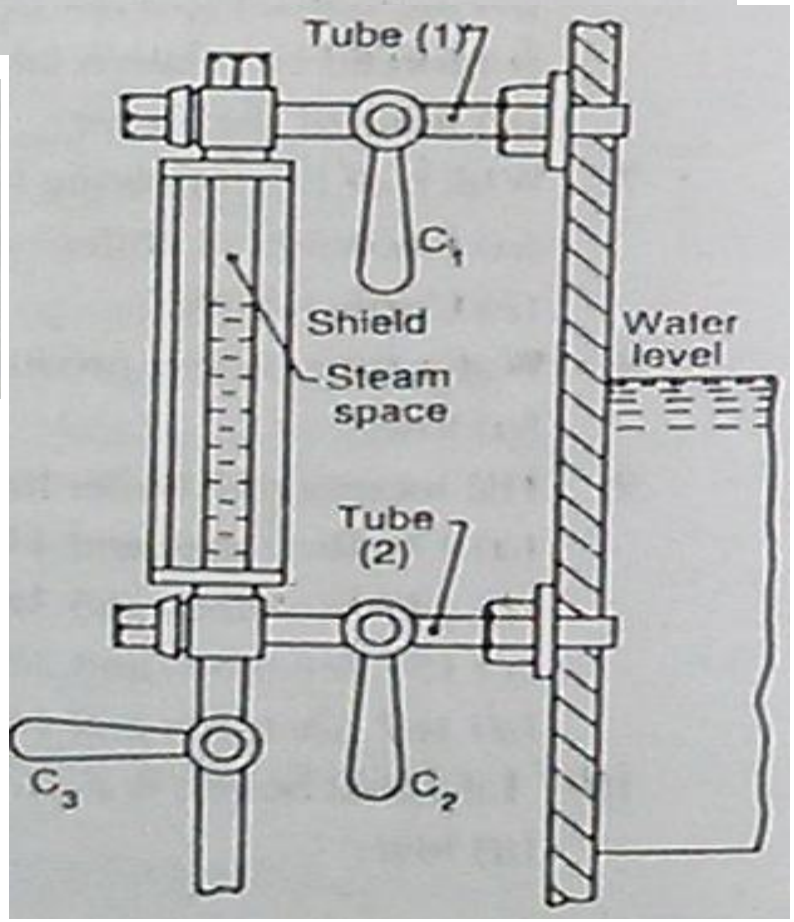
14.2. Boiler Mountings

These are the fittings, which are mounted on the boiler for its proper and safe functioning. Though there are many types of boiler mountings, yet the following are important from the subject point of view.

1. Water level indicator ; 2. Pressure gauge ; 3. Safety valves ; 4. Stop valve ; 5. Blow off cock ; 6. Feed check valve ; and 7. Fusible plug.

14.3. Water Level Indicator

It is an important fitting, which indicates the water level inside the boiler to an observer. It is a safety device, upon which the correct working of the boiler depends. This fitting may be seen in front of the boiler, and are generally two in number.



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14.4. Pressure Gauge

A pressure gauge is used to measure the pressure of the steam inside the steam boiler. It is fixed in front of the steam boiler. The pressure gauges generally used are of Bourden type.

gauge

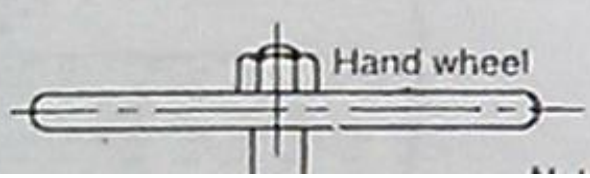
14.5. Safety Valves

These are the devices attached to the steam chest for preventing explosions due to excessive internal pressure of steam. A steam boiler is, usually, provided with two safety valves. These are

14.10. Steam Stop Valve

It is the largest valve on the steam boiler. It is, usually, fitted to the highest part of the shell by means of a flange as shown in Fig. 14.7. The principal functions of a stop valve are :

1. To control the flow of steam from the boiler to the main steam pipe.
2. To shut off the steam completely when required.



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14.11. Blow off Cock

The principal functions of a blow-off cock are :

1. To empty the boiler whenever required.
2. To discharge the mud, scale or sediments which are accumulated at the bottom of the boiler

The blow-off cock ...

14.12. Feed Check Valve

It is a non-return valve, fitted to a screwed spindle to regulate the lift. Its function is to regulate the supply of water, which is pumped into the boiler, by the feed pump. This valve must have its spindle lifted before the pump is started. It is

14.13. Fusible Plug

It is fitted to the crown plate of the furnace or the fire. Its object is to put off the fire in the furnace of the boiler when the level of water in the boiler falls to an unsafe limit, and thus avoids the explosion which may take place due to overheating of the furnace plate.

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14.14. Boiler Accessories

These are the devices which are used as integral parts of a boiler, and help in running efficiently. Though there are many types of boiler accessories, yet the following are important from the subject point of view :

1. Feet pump , 2. Superheater ; 3. Economiser ; and 4. Air preheater.

Fig. 14.11 shows the schematic diagram of a boiler plant with the above mentioned accessories.

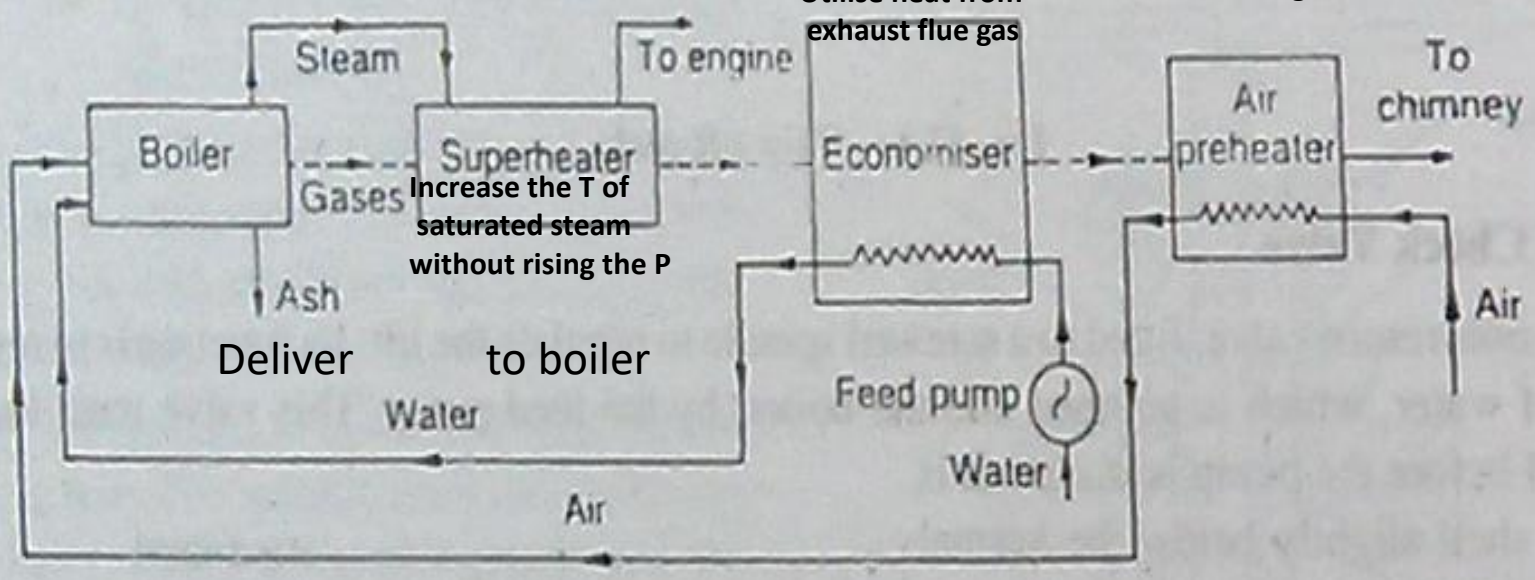


Fig. 14.11. Schematic diagram of a boiler plant.

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14.15. Feed Pump

We know that water, in a boiler, is continuously converted into steam, which is used by the engine. Thus we need a feed pump to deliver water to the boiler

14.16. Superheater

A superheater is an important device of a steam generating unit. Its purpose is to increase the temperature of saturated steam without raising its pressure. It is generally an integral part of a boiler,

14.17. Economiser

An economiser is a device used to heat feed water by utilising the heat in the exhaust flue gases before leaving through the chimney. As the name indicates, the economiser improves the economy of the steam boiler

14.18. Air Preheater

An air preheater is used to recover heat from the exhaust flue gases. It is installed between the economiser and the chimney. The air required for the purpose of combustion is drawn through the air preheater where its temperature is raised. It is then passed through ducts to the furnace. The air is passed through the tubes of the heater internally while the hot flue gases are passed over the outside of the tubes.

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15.1. Introduction

The performance of a steam boiler is measured in terms of its *evaporative capacity. However, the evaporative capacities of two boilers cannot be compared unless both the boilers have the same feed water temperature, working pressure, fuel and the final condition of steam. In actual practice,

15.2. Equivalent Evaporation

It is the amount of water evaporated from feed water at 100° C and formed into dry and saturated steam at 100° C at normal atmospheric pressure. It is, usually, written as "*from and at 100°C*".

As the water is already at the boiling temperature, it requires only latent heat at 1.013 bar to convert it into steam at the temperature (100° C). The value of this latent heat is taken as 2257 kJ/kg.

Mathematically,

Equivalent evaporation "from and at 100° C",

$$E = \frac{\text{Total heat required to evaporate feed water}}{2257}$$

Let t_1 = Temperature of feed water in °C,

h_{f1} = Enthalpy or sensible heat of feed water in kJ/kg of steam corresponding to t_1 °C (from steam tables),

h = Enthalpy or total heat of steam in kJ/kg of steam corresponding to a given working pressure (from steam tables),

$= h_f + x\, h_{fg}$ (For wet steam)

$= h_f + h_{fg} = h_g$ (For dry wet steam)

...or steam produced in kg/h.

$= h_g + c_p (t_{sup} - t)$ ... (For superheated steam)

m_e = Mass of water actually evaporated or steam produced in kg/h or kg/kg of fuel burnt.

We know that heat required to evaporate 1 kg of water

$= h - h_{f1}$

∴ Total heat required to evaporate m_e kg of water

$= m_e (h - h_{f1})$

and equivalent evaporation 'from and at 100° C"

$$E = \frac{m_e (h - h_{f1})}{2257}$$

Note : The factor $\frac{h - h_{f1}}{2257}$ is known as factor of evaporation, and is usually denoted by F_e. Its value is always greater than unity for all boilers.

15.3. Boiler Efficiency

It may be defined as the ratio of heat actually used in producing the steam to the heat liberated in the furnace. It is also known as thermal efficiency of the boiler. Mathematically,

Boiler efficiency or thermal efficiency,

$$\eta = \frac{\text{Heat actually used in producing steam}}{\text{Heat liberated in the furnace}} = \frac{m_e (h - h_{f1})}{C}$$

where m_e = Mass of water actually evaporated or actual evaporation in kg / kg of fuel, and

C = Calorific value of fuel in kJ/kg of fuel.

Notes : 1. If m_s = Total mass of water evaporated into steam in kg

and m_f = Mass of fuel used in kg

Then $m_e = \frac{m_s}{m_f}$ kg/kg of fuel

and $\eta = \frac{m_s (h - h_{f1})}{m_f \times C}$

2. If a boiler consisting of an economiser and superheater is considered to be a single unit, then the efficiency is termed as overall efficiency of the boiler.

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Example 15.1. *A boiler evaporates 3.6 kg of water per kg of coal into dry saturated steam at 10 bar. The temperature of feed water is 32° C. Find the equivalent evaporation "from and at 100° C" as well as the factor of evaporation.*

Solution. Given : $m_e = 3.6$ kg / kg of coal ; $p = 10$ bar ; $t_1 = 32°$ C

Equivalent evaporation 'from and at 100° C'

From steam tables, corresponding to a feed water temperature of 32° C, we find that

$$h_{f1} = 134 \text{ kJ/kg}$$

and corresponding to a steam pressure of 10 bar, we find that

$$h = h_g = 2776.2 \text{ kJ/kg} \quad \text{... (For dry saturated steam)}$$

We know that equivalent evaporation 'from and at 100° C',

$$E = \frac{m_e (h - h_{f1})}{2257} = \frac{3.6 (2776.2 - 134)}{2257} = 4.2 \text{ kg / kg of coal Ans.}$$

Factor of evaporation

We know that factor of evaporation

$$= \frac{h - h_{f1}}{2257} = \frac{2776.2 - 134}{2257} = 1.17 \text{ Ans.}$$

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Example 15.2. *The following observations were made in a boiler trial*

Coal used 250 kg of calorific value 29 800 kJ/kg, water evaporated 2000 kg, steam pressure 11.5 bar, dryness fraction of steam 0.95 and feed water temperature 34° C.

Calculate the equivalent evaporation "from and at 100° C" per kg of coal and the efficiency of the boiler.

Solution. Given : $m_f = 250$ kg , $C = 29\,800$ kJ/kg , $m_s = 2000$ kg , $p = 11.5$ bar ; $x = 0.95$, $t_1 = 34°$ C

Equivalent evaporation 'from and at 100° C'

From steam tables, corresponding to a feed water temperature of 34° C, we find that

$$h_{f1} = 142.4 \text{ kJ/kg}$$

and corresponding to a steam pressure of 11.5 bar, we find that

$$h_f = 790 \text{ kJ/kg} , \quad h_{fg} = 1991.4 \text{ kJ/kg}$$

We know that enthalpy or total heat of steam,

$$h = h_f + x\, h_{fg} = 790 + 0.95 \times 1991.4 = 2681.8 \text{ kJ/kg}$$

and mass of water evaporated per kg of coal

$$m_e = m_s / m_f = 2000 / 250 = 8 \text{ kg / kg of coal}$$

∴ Equivalent evaporation 'from and at 100° C'

$$E = \frac{m_e (h - h_{f1})}{2257} = \frac{8\,(2681.8 - 142.4)}{2257} = 0.682 \text{ kg/kg of coal}$$ **Ans.**

Efficiency of the boiler

We know that efficiency of the boiler,

$$\eta = \frac{m_e (h - h_{f1})}{C} = \frac{8\,(2681.8 - 142.4)}{29\,800} = 0.682 \text{ or } 68.2\ \%$$ **Ans.**

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Please study
-Chapter (7, 8) and (13, 14, 15) for week one and two,
-Chapter 10 for week three,
-Chapter 6 and 26 for week four,
-Chapter 25 for week five,
These are the syllabus for mid term.

Thank you for your attention